

Effects of Experiential Priming in Virtual Reality on Flavor Profiles of Real-World Beverages

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ABSTRACT

Earlier research establishes the predominance of virtual over real textures, flavors, and scents when two different stimuli are presented to users. However, the effects of virtual priming, rather than simultaneous exposure, has not been thoroughly investigated. Experiential priming involves the presentation of one set of stimuli prior to a second set of stimuli. We examined whether experiential priming conducted in a virtual environment influences real-world flavor profile perception. The results of our preliminary work suggest that VR experiential priming produces perceptual differences in the real world and warrants further investigation.

Index Terms: Human-centered computing—Flavor perception—Experiential priming

1 INTRODUCTION & BACKGROUND

Using virtual reality to influence our experience and flavor profiles of the real world has the potential to help individuals who have dietary restrictions, harmful addictions, pica, or as a unique way to experience food and drink. Virtual Environments could be used to manipulate the taste perception of what a user is actually ingesting. Prior studies provide some indication that external stimuli influence gustatory perception. Ramos-de-la-Peña et al. describe creating seven virtual environments, each intended to elicit a different feeling in the participant [5]. Scenes associated with hot and cold temperatures altered perception of bitterness, moisture, sweetness, and spiciness in ketchup and salad dressing [5]. Niki et al. investigated whether virtually displaying strawberry sponge cake and orange juice in the place of bitter flavored consumables would impact the flavors perceived by participants [4]. This study found evidence that this technique increased perceived sweetness and decreased the time a bitter flavor would persist. Huang et al. conducted a study consisting of three experiments involving red and green Chinese tea and determined virtual tea color influenced perceived saltiness of sampled teas [2]. Wang Qian et al. investigated whether changing the appearance of coffee in VR to make it look milkier could alter its perceived flavor or participants' enjoyment of the drink, and concluded that virtual color significantly influenced perceived creaminess [6]. In a study conducted by Ammann et al., 30% of participants identified the primary flavor of lemon cake as chocolate when it was colored brown, showing that visual expectations can drastically change perceived taste [1]. Through our experiment, we sought to determine if virtual Experiential Priming has an effect on the taste or flavor profiles of a real-world beverage. We hypothesized



Figure 1: A Participant's View of Cutting Lemons.

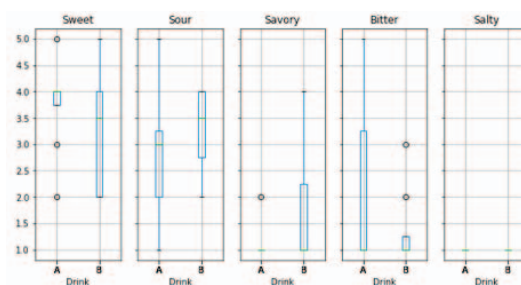


Figure 2: Perceived Flavor Profiles of Real-World Beverage Condition, where A represents strawberry lemonade and B is regular lemonade demonstrating the distinct differences that a participant should detect.

that having this type of experience with virtual food, in this case mixing a beverage, that the flavor profile of the real beverage would be influenced by that virtual experience.

2 METHODS

Prior to the experiment, participants read and signed a consent form describing the procedure outlined below. Participants put on a Vive headset and received basic instruction on using the controllers and following prompts in the virtual environment to slice lemons, add sugar and water to a glass, and blend strawberries, combining the products to prepare a virtual strawberry lemonade beverage. After this priming condition, the researcher then served a participant a beverage in a covered container with an opaque straw, omitting smell. The participant completed a mid-survey questionnaire regarding the flavor profile of the beverage. We ask participants to provide a description of the beverage, as well as rank various flavor components (sweet, salty, etc.) on a five-point scale. The researcher then presented the participant with two more beverages in succession: strawberry lemonade and lemonade. We randomized the first beverage to minimize the statistical biasing effects of the original beverage presented. Each participant completed a final questionnaire regarding the flavors of the last two beverages and their similarity to or differences from the virtual strawberry lemonade and the previous drink tasted.

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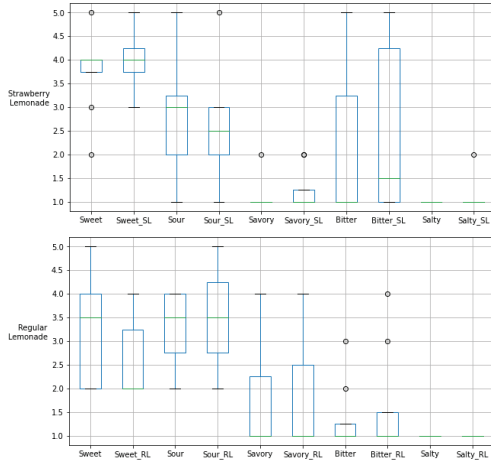


Figure 3: Reported Flavor Profiles of Identical Beverages

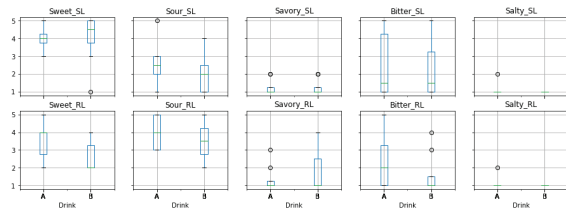


Figure 4: Reported Flavor Profiles of Real-World Beverages

3 RESULTS

We obtained preliminary results from a sample size of $n = 16$ participants (men $N=12$, women $N=4$) with ages ranging from 18 to 35. Participants reported VR Experience ($M=3.69$, $SD=1.96$) and VR Use Frequency ($M=3.13$, $SD=1.82$) on a scale from 1 to 7. We randomly assigned strawberry lemonade or regular lemonade to participants following the experiential priming stage. A flavor profile comparison graph is included in figure 2. The flavor profiles of the original beverage and the taste-test comparison beverage are provided in figure 3, grouped by randomized beverage. The first label denotes the flavor profile of the original beverage, and labels with subscripts represent the flavors in the comparison beverage.

We asked the participants the extent (on a scale of 1=not related to 7= closely related) to which they thought the virtual beverage resembled the real beverage. Strawberry lemonade was reported more closely to the given real beverage ($M=4.75$, $SD=1.91$) than regular lemonade ($M=4.5$, $SD=2.45$).

When presented with both beverages in a random order to minimize the effects of sequence, participants provided the following ratings for the flavors present in each beverage, grouped by the beverage participants originally sampled, where A is strawberry lemonade and B is regular lemonade.

Table 1 tabulates what beverages participants were originally given, which comparison beverage they identified as more similar to the beverage seen in VR, and which beverage they identified as the first beverage they tasted. SL represents strawberry lemonade, RL represents regular lemonade, and ND represents "neither drink".

Participants commented on the differences between the virtual and real beverage. Notably, one participant said, "Didn't get much of the strawberry" after consuming strawberry lemonade, while another explicitly stated, "I felt like I was drinking a lemonade/strawberry drink" after consuming regular lemonade. Following the presenta-

Table 1: Beverage Similarity Perception

Given	VR beverage			Real beverage		
	SL	RL	ND	SL	RL	ND
SL	5	1	2	5	2	0
RL	3	1	2	0	5	1

tion of both regular and strawberry lemonade, we asked participants for further descriptions of both beverages and on the experiment as a whole. Many identified the drinks correctly, but some perceived "fruit juice", "lemonade" when presented with strawberry lemonade, and one expressed, "I wasn't [sic] sure if I was drinking lemonade or strawberry lemonade." when given regular lemonade.

4 DISCUSSION AND CONCLUSION

Through our experiment, we sought to determine if virtual Experiential Priming has an effect on the taste or flavor profiles of a real-world beverage. Analyzing the data from the reports after the first taste reveal that participants incorrectly identified regular lemonade as significantly more closely related to strawberry lemonade. This is interesting since actual strawberry lemonade was included as one of the real-world beverage conditions. In the follow up final comparison of both beverages, participants were able to distinguish the two beverages from each other and be able to correctly identify the beverage types, even though they incorrectly identified the real-world beverage before. These results suggest that virtual Experiential Priming has some effect on how we perceive real-world beverages, though it is not clear to what extent this priming has. In a prior study, participants reported changes in perceived flavor profiles based on visual color change [3]. While our study additionally finds that the change in flavor perception persists after VR exposure (rather than during in the prior study), a first step to determine the effect of experiential priming as compared to visual only, but further investigation is needed to determine if the virtual information is affecting the flavor profile or the act of mixing the drink itself (experiential priming).

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